

Amanda Zhang

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EDUCATION

University of California, Berkeley

May 2027

Bachelor of Science in Electrical Engineering & Computer Sciences

GPA: 3.71/4.00

Relevant Coursework: Optimization Models, Efficient Algorithms, Computer Security, Computer Architecture, Data Structures, Discrete Math & Probability, Circuit Analysis

Organizations: Computer Science Mentors, Society of Women Engineers, Go Club, Cal Climbing

EXPERIENCE

Theo AI @ Async Labs

June 2025 – August 2025

Software Engineering Intern

- Built and shipped an end-to-end RAG pipeline using LlamaCloud + LLM agents to retrieve and summarize 1000+ historical IT tickets, cutting troubleshooting latency through automated similarity search and next-step generation.
- Designed AI agent architectures for autonomous workstation management; compared ReACT vs computer-use agents and authored 30+ production-ready system-admin automation scripts.

UC Berkeley Electrical Engineering & Computer Sciences

June 2025 – Current

Undergraduate Tutor (UCSI) for Data Structures and Algorithms (CS 61B)

- Supporting 1200+ students per semester in office hours, small group tutoring, and online discussion forum.
- Facilitating weekly small group sections, delivering engaging mini-lectures and interactive lessons on data structures and algorithms concepts including trees, graphs, sorting algorithms, runtime analysis.
- Collaborating with the course staff team to shape curriculum and maintain UCB course standards through exam development/grading and curriculum planning to maximize student learning and course experience.

Space Cookies, FIRST Robotics Team & Girl Scouts Troop

August 2020 – June 2024

Software Captain

- Led 20-person software team in developing autonomous subsystems and navigation, control systems, and code framework for competitive robotics using Java and GitHub version control.
- Designed a 10-week technical workshop curriculum covering Java fundamentals and robotics software development.
- 2023 Regional Innovation in Controls Award; 2x Regional Impact Award (Most Prestigious FIRST Award).

UC Berkeley Hybrid Robotics Lab

June 2023 – August 2023

Research Intern with Professor Koushil Sreenath

- Implemented reinforcement learning locomotion models (proportional-derivative control) in PyTorch + Mujoco to optimize robot movement.
- Analyzed how environmental parameters affected sim-to-real policy transfer for quadrupedal robot locomotion.

PROJECTS

BetaBot: AlphaGo's Beta Counterpart – Python, Deep Reinforcement Learning, Amazon Web Services

2026

- Building a Go-playing AI from scratch, including a board-state tracking tree and a neural network for move prediction.
- Designing a reinforcement learning training pipeline to continuously improve model consistency and strength.
- Developing deployment infrastructure to Online Go Server (OGS), and a teaching interface for students to play and review games with the bot.

Cal Hiking and Outdoors Society (CHAOS) Gear Shed App – [Next.js](#), Supabase (DB + Auth)

2026

- Developing a mobile web app to reliably and simply manage CHAOS club gear checkout and returns.
- Designing and maintaining an easily editable database of gear items and tracking real-time status of items.
- Implementing a QR code system to streamline and standardize the gear check-in and out process.

Scheme Interpreter – Python

2024

- Built a Python-based interpreter for a subset of the Scheme language, implementing evaluation of primitive operations, function calls, and nested list expressions.
- Designed lexical parsing and evaluation logic to recursively interpret symbolic expressions.

SKILLS

Computer Languages: Python, Java, C, Bash, SQL, RISC-V

Frameworks & Tools: React, Flask/FastAPI, Docker, Linux, Git/ GitHub

AI/ML: RAG pipelines, vector databases (LlamaCloud), LLM integration, LangGraph

Languages: English (fluent), Mandarin Chinese (fluent)